

Remark 1 (Two categories versus one). *The forgetful functor from factorization modules (ii) to fibers (i) forgets the flat spectral connection. The R -matrix is the holonomy datum that reconstructs the family from its fibers. Within the E_∞ -chiral class, the pole-free commutative algebras have $R(z) = \tau$: the flat connection is trivial and the two module categories collapse. For E_∞ -chiral algebras with OPE poles (vertex algebras such as $\widehat{\mathfrak{g}}_k$ or Vir_c), $R(z) \neq \tau$ but is derived from the local OPE, so the two categories are related by an explicit twist. For genuine E_1 -chiral algebras, the R -matrix is independent input — the nontrivial datum distinguishing the two categories. This is why the spectral parameter and quantum group structure are invisible in the pole-free commutative limit and progressively more visible as one moves through the three tiers.*

Here $\kappa = c/2$ is the *intrinsic* curvature of Vir_c —the matter sector alone. The full gravitational generating function uses instead the *effective* curvature $\kappa_{\text{eff}} = (c - 26)/2$, which incorporates the ghost contribution. Explicitly, with $\kappa = c/2$:

Proof. The MC element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ satisfies the MC equation because $D_{\mathcal{A}}^2 = 0$. At the level of coderivations, therefore, $\sigma(\Theta_{\mathcal{A}}) = D_{\mathcal{A}^!} - d_0 = \Theta_{\mathcal{A}^!}$. The sign arises when one passes to genus- g projections through the bilinear pairing. The involution σ acts anti-symmetrically on the pairing: $\langle \sigma(x), \sigma(y) \rangle = -\langle x, y \rangle$. The projection π_g^* extracts the component in Hodge degree g via this pairing, so the anti-symmetry contributes an overall sign:

$$\sigma \circ \pi_g^*(\Theta_{\mathcal{A}}) = -\pi_g^*(D_{\mathcal{A}^!} - d_0) = -\pi_g^*(\Theta_{\mathcal{A}^!}).$$

This is the complementarity relation $\pi_g^*(\Theta_{\mathcal{A}}) + \pi_g^*(\Theta_{\mathcal{A}^!}) = 0$ at every genus. $\square$

Theorem 2 (Test; Heuristic). *Statement.*

Evidence. The argument is heuristic because it assumes something. $\square$